

NASSAU CANDY DISTRIBUTOR

Product Line Profitability & Margin Performance

Research Paper: EDA, Insights & Strategic Recommendations

Data Period: January 2024 – December 2025 | 10,194 Orders | 15 Products | 3 Divisions | 4 Regions

1. Executive Overview

Nassau Candy Distributor operates across three product divisions — Chocolate, Sugar, and Other — serving four US regions over a 24-month period. This research paper presents an exploratory data analysis (EDA) of 10,194 transaction records, identifies key profitability drivers and risk areas, and offers actionable strategic recommendations.

2. Dataset Description & Methodology

2.1 Dataset Structure

The dataset contains 18 columns including: Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Country/Region, City, State/Province, Postal Code, Division, Region, Product ID, Product Name, Sales, Units, Gross Profit, and Cost.

2.2 Derived Metrics

- Gross Margin (%) = $\text{Gross Profit} / \text{Sales} \times 100$
- Profit per Unit = $\text{Gross Profit} / \text{Units}$
- Revenue Share (%) = $\text{Product Revenue} / \text{Total Revenue} \times 100$
- Profit Share (%) = $\text{Product Profit} / \text{Total Profit} \times 100$
- Cumulative Revenue % for Pareto analysis

2.3 Scope

Analysis covers January 2024 through December 2025 with no missing values identified in key financial fields. Date parsing used day-first format consistent with the source data encoding.

3. Key Performance Indicators

KPI	Value
Total Revenue	\$141,783.63
Total Gross Profit	\$93,442.80
Overall Gross Margin	65.9%
Total Units Sold	38,654
Total Orders	10,194
Number of Products	15
Number of Divisions	3
Number of Regions	4

Nassau Candy achieves an impressive overall gross margin of 65.9%, indicating strong pricing power and cost discipline at the portfolio level. However, this headline figure masks significant variation at the product level.

4. Division-Level Performance

Division	Revenue (\$)	Gross Profit (\$)	Margin (%)	Units	Rev Share (%)
Chocolate	\$131,692.90	\$88,824.62	67.4%	37,275	92.9%
Other	\$9,663.25	\$4,333.45	44.8%	1,242	6.8%
Sugar	\$427.48	\$284.73	66.6%	137	0.3%

Chocolate is the overwhelming revenue engine, contributing 92.9% of total revenue and 95.1% of gross profit. Its 67.4% margin is the highest among divisions. The Other division, while smaller, suffers from a significantly lower 44.8% margin, dragged down by the underperforming Kazookles product. The Sugar division is niche but margin-efficient at 66.6%.

5. Product-Level EDA

5.1 Gross Margin by Product

Product Name	Division	Revenue (\$)	Gross Profit (\$)	Margin (%)	Profit/Unit (\$)
Wonka Bar - Scrumdiddlyumptious	Chocolate	\$27,874.80	\$19,357.50	69.4%	\$2.50

Product Name	Division	Revenue (\$)	Gross Profit (\$)	Margin (%)	Profit/Unit (\$)
Wonka Bar - Triple Dazzle Caramel	Chocolate	\$28,485.00	\$18,610.20	65.3%	\$2.45
Wonka Bar - Milk Chocolate	Chocolate	\$26,867.75	\$17,443.37	64.9%	\$2.11
Wonka Bar - Nutty Crunch Surprise	Chocolate	\$23,574.95	\$16,819.95	71.3%	\$2.49
Wonka Bar - Fudge Mallows	Chocolate	\$24,890.40	\$16,593.60	66.7%	\$2.40
Lickable Wallpaper	Other	\$7,860.00	\$3,930.00	50.0%	\$10.00
Wonka Gum	Other	\$597.50	\$310.70	52.0%	\$0.65
Everlasting Gobstopper	Sugar	\$130.00	\$104.00	80.0%	\$8.00
Hair Toffee	Sugar	\$76.50	\$59.50	77.8%	\$3.50
Fizzy Lifting Drinks	Sugar	\$78.75	\$47.25	60.0%	\$2.25
Laffy Taffy	Sugar	\$53.73	\$33.48	62.3%	\$1.24
SweeTARTS	Sugar	\$46.7%	\$28.70	46.7%	\$0.70
Nerds	Sugar	\$15.00	\$7.00	46.7%	\$0.70
Fun Dip	Sugar	\$12.00	\$4.80	40.0%	\$0.60
Kazookles	Other	\$1,205.75	\$92.75	7.7%	\$0.25

5.2 Pareto Concentration

The top 5 products are all Wonka Bar variants and together account for approximately 93.4% of total revenue and 95.1% of total gross profit. This extreme concentration represents both a strength and a strategic risk. The bottom 10 products contribute less than 7% of revenue combined, signaling an opportunity for portfolio rationalization.

5.3 Critical Risk: Kazookles

Kazookles generates only 7.7% gross margin — the lowest in the portfolio by a wide margin. On \$1,205.75 in revenue it generates just \$92.75 in profit. With 371 units sold it is the third-highest volume product in the Other division, yet barely breaks even. This product warrants immediate review for repricing, cost renegotiation, or discontinuation.

6. Regional Performance

Region	Revenue (\$)	Gross Profit (\$)	Margin (%)
Pacific	\$46,301.53	\$30,485.94	65.8%

Region	Revenue (\$)	Gross Profit (\$)	Margin (%)
Atlantic	\$41,197.24	\$26,973.70	65.5%
Interior	\$32,037.60	\$21,282.49	66.4%
Gulf	\$22,247.26	\$14,700.67	66.1%

All four regions operate within a tight 65.5–66.4% gross margin band, confirming that profitability differences are driven by product mix rather than regional pricing or cost dynamics. Pacific leads in revenue at \$46,301, while Gulf is smallest at \$22,247. Regional margin uniformity is a sign of consistent pricing execution.

7. Temporal Trends

Monthly revenue analysis reveals a consistent seasonal pattern across both years:

- January–February: slowest months (~\$1,400–\$3,500/month)
- September, November, December: consistently peak months, exceeding \$7,000–12,000/month
- Overall revenue trend is upward: December 2025 (\$12,338) is nearly 6× January 2024 (\$2,094)
- Gross margins remained stable throughout, showing no seasonal compression

The accelerating revenue trend in H2 2025 suggests either market expansion, increased distribution, or successful product push campaigns. This trend should be maintained with inventory and logistics planning aligned to Q4 peaks.

8. Key Insights

8.1 Concentration Risk

92.9% revenue dependency on one division (Chocolate) and the top 5 SKUs accounting for ~93% of revenue creates fragility. Any supply disruption, margin compression, or competitive pressure on Wonka Bars would have outsized impact on total business performance.

8.2 High-Value Niche Products

Everlasting Gobstopper (80% margin, \$8.00 profit/unit) and Hair Toffee (77.8%, \$3.50/unit) are extremely efficient profit contributors despite negligible revenue. These products deserve targeted scale-up investment.

8.3 Kazookles: Near-Zero Profitability

At 7.7% margin with 371 units sold, Kazookles consumes sales resources and shelf space for almost no return. It is the clearest candidate for either a major reprice or discontinuation.

8.4 Sugar Division Scale Opportunity

The Sugar division averages ~66% margins but generates only \$427 in revenue. This implies either very limited distribution, low demand, or nascent product launches. A targeted push could yield disproportionate profit given the existing margin structure.

9. Strategic Recommendations

Priority	Action	Target	Expected Outcome
1 – CRITICAL	Discontinue or reprice Kazookles	Other Division	Eliminate 7.7% margin drag
2 – HIGH	Scale Everlasting Gobstopper & Hair Toffee	Sugar Division	+\$5K–10K in high-margin profit
3 – HIGH	Diversify beyond Chocolate division	Portfolio	Reduce concentration risk
4 – MEDIUM	Negotiate costs for Fun Dip, Nerds, SweetARTS	Sugar/Other	Push margins above 50%
5 – MEDIUM	Invest in Gulf region expansion	Regional	Close revenue gap vs. Pacific
6 – LOW	Align inventory build with Q4 seasonal peaks	Operations	Avoid stockouts in Nov–Dec

10. Conclusion

Nassau Candy Distributor operates with a strong 65.9% gross margin, underpinned by a powerful Chocolate division and the Wonka Bar franchise. The strategic challenge is reducing SKU-level risk, expanding high-margin niche products, and addressing the Kazookles margin crisis. Regional performance is uniformly healthy, suggesting the core business model is sound and scalable.

The 2024–2025 revenue growth trend — particularly the accelerating Q4 peaks — provides a positive basis for investment in expansion. Prioritizing margin improvement in the Other division and deliberate scaling of the Sugar division would materially strengthen portfolio resilience and long-term profitability.